Robotics and Autonomous Systems 97 (2017) 182–192

Contents lists available at ScienceDirect

Robotics and Autonomous Systems

journal homepage: www.elsevier.com/locate/robot

Needle in a haystack: Interactive surgical instrument recognition through perception and manipulation☆

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HIGHLIGHTS

- A machine vision to pick cluttered surgical instruments for a robotic scrub nurse.
- The recognition approach involves visual perception and dexterous manipulation.
- A force-based reactive grasping protocol is proposed.
- A new segmentation algorithm based on parts-based detection is presented.

ARTICLE INFO

Article history:
Received 7 September 2016
Received in revised form 27 July 2017
Accepted 25 August 2017
Available online 7 September 2017

ABSTRACT

This paper presents a solution to the challenge of accurate surgical instrument recognition by a Robotic Scrub Nurse (RSN) in the Operating Room (OR). Surgical instruments, placed on the surgical mayo tray, are often cluttered, occluded and display specular light which poses a challenge for conventional recognition algorithms. To tackle this problem we resort to a hybrid computer vision and robotic manipulation combined strategy. The instruments are first segmented and pose is estimated, then the RSN system picks up the unknown instruments and presents them to the optical sensor in the determined pose. Last, the instruments are recognized and delivered. Experiments were conducted to evaluate the performance of the proposed segmentation, grasping and recognition algorithms, respectively. The proposed patch-based segmentation algorithm can achieve an F-score of 0.90. The proposed force-based grasping protocol can achieve an average picking success rate of 92% with various instrument layouts, and the proposed attention-based instrument recognition module can reach a recognition accuracy of 95.6%. Experimental results indicate the applicability and effectiveness of a RSN to perform accurate and robust surgical instrument recognition.

1. Introduction

American hospitals are facing critical problems of lacking nurses. According to a recent report [1], there will be a shortage of 260,000 registered nurses by 2025 in the USA. Such shortage could lead to an increase in mortality. For example, it was reported that patient mortality risk is 6% higher in hospitals understaffed with nurses compared to units fully staffed [2]. One solution to the nurse shortage problem is to develop systems which can take over the highly mechanical, mundane and repetitive tasks from the nurses, so as to allow them to focus more on complex tasks. Among all the tasks that the nurses are responsible for, the surgical instrument preparation and delivery task is one of the most important and dominant cases, which the proposed Robotic Scrub Nurse (RSN) system is trying to address and take over.

To build a robust RSN system, accurate instrument recognition is a critical component. Although humans can perform such tasks elegantly with the help of both vision and tactile information, it is quite a challenging task for robots to accurately recognize them. The instruments are often clustered together on the mayo stand and have a reflective and uniform appearance in both shapes and colors, as shown in Fig. 1. To enable a RSN system to serve robustly in the Operating Room (OR), it is critical for the robot to recognize the instrument accurately, localize them precisely and manipulate them robustly. The aim of this paper is to propose one such system which meets the aforementioned requirements.

☆ Research supported by the NPRP award (NPRP 6-449-2-181) from the Qatar National Research Fund (a member of The Qatar Foundation). The statements made herein are solely the responsibility of the authors.

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http://dx.doi.org/10.1016/j.robot.2017.08.013

Fig. 1. A realistic mayo tray configuration.

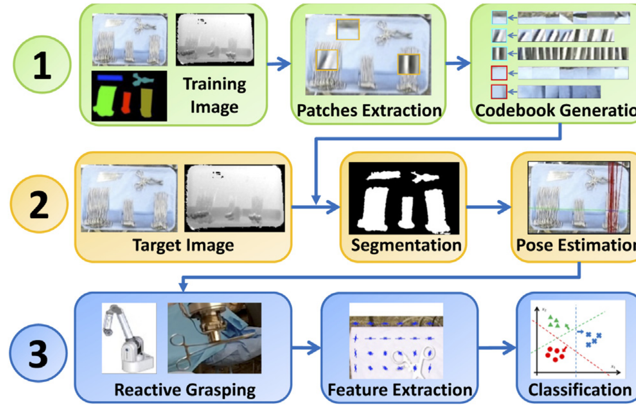

Fig. 2. System Architecture.

2. Related work

Robotic systems have been brought into the OR in mainly three different forms [3], (1) handheld robotic tools; (2) teleoperated surgical systems and (3) autonomous surgical assistants. The first two cases focus on designing robots to increase surgeon's sensorimotor capabilities. Such systems include the minimally invasive telesurgery system da Vinci [4], the steady-hand robotic system for microsurgical augmentation [5] and the touchless telerobotic surgery system which allows free hand gesture interactions [6]. The third category refers to robots assisting the surgeons and nurses without directly touching the patient. One specific type of robotic surgical assistant is the RSN system which manages, delivers and retrieves surgical instruments for surgeons upon request. Such RSN system can free the nurses to perform multiple tasks concurrently [7] and has the potential to reduce communication errors [8]. Many of the previous efforts in RSN development have been spent on the human–robot interaction part, where gestures [9,10], speech [11], haptics [12], EEG/EMG [13] and body-worn sensors [14] were used to enable communications between surgeons and RSN systems. However, there is a lack of attention to the problem of instrument recognition and manipulation in realistic/uncontrolled settings. For example, non-overlapping instrument on a specially-designed mayo platform are addressed [15]. Alternatively, the instrument locations were recorded ahead of time and their identities were recognized through infrared labels [12]. This work tried to bridge the gap of recognizing surgical instruments in an uncontrolled surgical setting without the usage of additional labels.

The proposed recognition algorithm consists of instrument segmentation, reactive grasping and instrument recognition. These problems are at the core of this paper and the relevant literature are discussed in the following.

Image segmentation often involves dividing the entire image into smaller coherent groups for further processing. The three major approaches are edge-based, region-based and threshold-based methods. The edge-based approach first finds edges on the image using various filters (e.g., wavelet transformation [16], Gabor filter [17]), and then links the adjacent edges to form a contour (e.g., K-means [18], Fuzzy C-means [19]). However, they are sensitive to noise and do not work well on images with smooth transitions and low contrast. On the other hand, the region based approach consider the grayscale similarities in neighboring pixels to merge and split them into sub-regions, e.g., Watershed [20], Chan–Vese model [21], superpixel [22], GrabCut [23]. Some of the region-based algorithms require initial seed region to start the segmentation (by human or using some heuristic). The last group, threshold-based method, compares the image intensity levels with a threshold, which are found automatically using Otsu's [24], genetic algorithms [25] or particle swarm optimization [26], to mention a few. More recently, Convolutional Neural Network (CNN) algorithms were used to achieve both segmentation and recognition with satisfactory accuracies [27,28]. However, these algorithms have a large memory foot-print and an ambitious hardware requirement for usage. The proposed patch-based segmentation algorithm solves both segmentation and recognition at the same time, and does not require such hardware.

The instrument recognition algorithm follows an interactive recognition strategy, where the object-of-interest is grasped, manipulated and observed actively by a robot to determine its identity. Lyubova et al. proposed a framework where a humanoid robot is able to recognize unknown objects through interaction and observation [29]. Such procedure mimics the process of human learning, and increases the recognition accuracy by cumulating the confidences from multiple modalities (i.e., vision, tactile and affordance [30]). However, to use such approach, the robot is required to be able to grasp unknown objects successfully, which on the other hand usually requires accurate object identity and properties, such as their 3D models [31] or physical properties [32]. This "chicken-and-egg" problem can be solved through reactive grasping protocols, which do not require full object models, and are at the core of our work. Some reactive grasping algorithms leverage tactile feedback [33] and optical proximity sensors [34] to grasp the unknown object robustly. The proposed force-based reactive grasping protocol also works without full object models.

This paper makes the following contributions: (1) a segmentation algorithm based on visual codebook generation and weighted histogram backprojection; (2) a force-based reactive grasping protocol to enable reliable instrument grasping; (3) a surgical instrument recognition algorithm. All these components are key for the successful implementation of a RSN.

3. System architecture

Recognition and grasping of surgical instruments from mayo trays in a clinical setting is challenging due to: (a) the instruments are packed together leading to occlusion, discontinuities and clutter; (b) salient points are difficult to find on metallic instruments due to their uniform and reflective composition.

We tackled this challenge through an active recognition process, which involves manipulation and then recognition. The system architecture is shown in Fig. 2. Firstly, a codebook of local appearances is created in the training phase (step #1). During usage, the packs of instruments are segmented and their poses

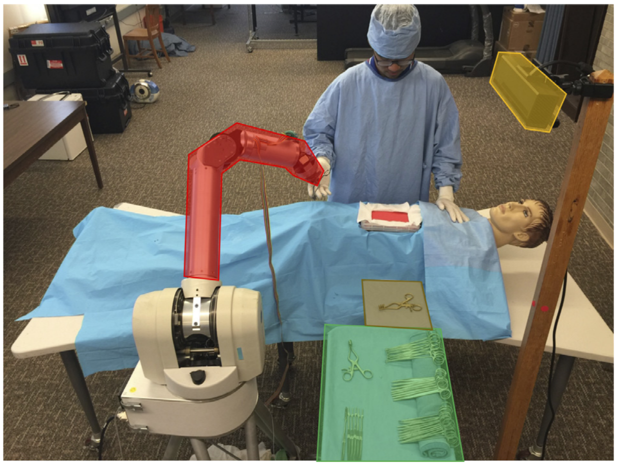

Fig. 3. Illustration of the system. The highlighted regions are mayo tray (green), recognition pad (brown), Kinect (yellow) and robotic arm (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

are estimated (step #2). Finally, a robotic arm grasps the surgical instrument and places it over a side platform where the recognition is performed (step #3). Once the instrument is recognized, clusters of instruments are assigned a specific label. An illustration of the developed system is given in Fig. 3.

4. Methodology

The following section describes the instrument segmentation, reactive grasping and recognition algorithms respectively.

4.1. Instrument pack segmentation

To enable the robot to pick up and manipulate surgical instruments, each instrument needs to be localized on the mayo tray first. The instrument segmentation process consists of: (1) segmenting mayo stand using adaptive threshold; (2) a training process where color codebooks of distinctive instrument appearances were built; (3) a segmentation process leveraged on weighted histogram back-projection using the trained color codebook; (4) pose estimation for each cluster of instruments. Each step is discussed in the following.

4.1.1. Mayo segmentation

The setup consists of a Kinect sensor placed directly on top of the mayo stand facing down to capture a complete view of the region of interest (as shown in Fig. 3). A Kinect sensor delivers a color image I_C and depth image I_D, which serve as the input to the mayo and instrument segmentation process.

Otsu's threshold [24] and contour analysis are used together to get the mask of the mayo tray (denoted as M_{mayo}). Otsu's adaptive threshold was applied on the raw depth image I_D to generate a foreground mask M_{fg}. The *rectangularity* was then computed for every contours in M_{fg}. The *rectangularity* β is defined as the ratio of the width to the height of the rotated bounding box of the contour. A standard mayo stand has a rectangular shape which can be identified by choosing the contour in the foreground M_{fg} which has the largest *rectangularity* among the candidates. This process is depicted in Fig. 4.

4.1.2. Codebook generation

From the mayo stand image, each pile of unknown instrument needs to be segmented and their poses must be determined. The first step is to segment the foreground (instruments) from the background (mayo surface). A classification-based segmentation technique was utilized to accomplish this task. Classification-based segmentation consists of building a class-specific codebook of local appearance (both color and texture) and then classifying each pixel in the image as belonging to one of those classes. Suppose there are R classes in the image, whose labels are denoted as $c_1, \ldots, c_R$. First, a class-specific codebook of local appearances, denoted as ϕ^r, was built for each object category c_r. The codebook ϕ^r consists of K characteristic local color models, represented by color histograms ($hist_k^r$), and a corresponding weight factor (w_k^r), as shown below:

$$\phi^r = \left\{\phi_k^r\right\} = \left\{\left(w_k^r, hist_k^r\right) | k = 1, \ldots, K\right\} \tag{1}$$

where w_k^r and $hist_k^r$ represents the k_{th}weight and local color model for codebook ϕ^r. The histograms $hist_k^r$ are used later to generate a back-projected image for segmentation, and the weights w_k^r represent the relative importance of each color model. To build the codebook, we proposed a variant of the *Bag of Visual Words* (BoVW) algorithm [35]. The traditional BoVW method fails when directly used for instrument recognition, due to the similarity of the appearances when cluttered together. Therefore, we adopted an active recognition approach where instrument packs were segmented using BoVW as whole fuzzy regions. After that, the pose for each instrument pack was estimated and a robotic arm was used to change the instrument pose for maximum visibility for further recognition.

In the codebook generation process, N random patches $P_i\ (i = 1 \ldots N)$ were extracted from manually segmented image regions of each category c_r, from each training image. Each patch P_i is associated with local color and depth information and a patch label θ_i, all together represented as:

$$P_i = \{\theta_i, \{I_C, I_D\} \cap W(x_i, y_i)\}, i = 1 \ldots N \tag{2}$$

where i is the index of the patch and $\theta_i \in \{c_1, \ldots, c_R\}$ is the label of the patch. $\{I_C, I_D\}$ represents the raw color and depth image, and $W(x_i, y_i)$ is a squared window of size $w \times w$ centered at (x_i, y_i), which is the location of the patch P_i. From each patch P_i, a stack of color and depth features were extracted, forming a feature representation F_i for patch P_i. The feature set F_i includes (with d as dimensions):

- Histogram of grayscale pixel values (d = 16).
- Histogram of Hue and Saturation channel pixel values (d = 32).
- Histogram of oriented gradient (HOG) from the grayscale image (d = 36).
- Global descriptor of grayscale image: mean and standard deviation of pixel values, and mean of the Laplacian image (d = 3).
- Histogram of raw depth image pixels (d = 16).
- Global descriptor of raw depth image: mean and standard deviation of pixel values, and mean of the Laplacian image (d = 3).

In total, the feature representation F_i has a dimension of 106. The set of all the patches for category c_r were clustered using their feature representation F_i with Gaussian Mixture Models (GMM):

$$\{F_i|\theta_i = c_r\} = \sum_{k=1}^{K} \alpha_k^r \mathcal{N}\left(F|\mu_k^r, \Sigma_k^r\right) \tag{3}$$

where each cluster center $\left(\mu_k^r, \Sigma_k^r\right)$ represents the k_{th} prototypical local feature appearance for object type c_r, and the weight α_k^r

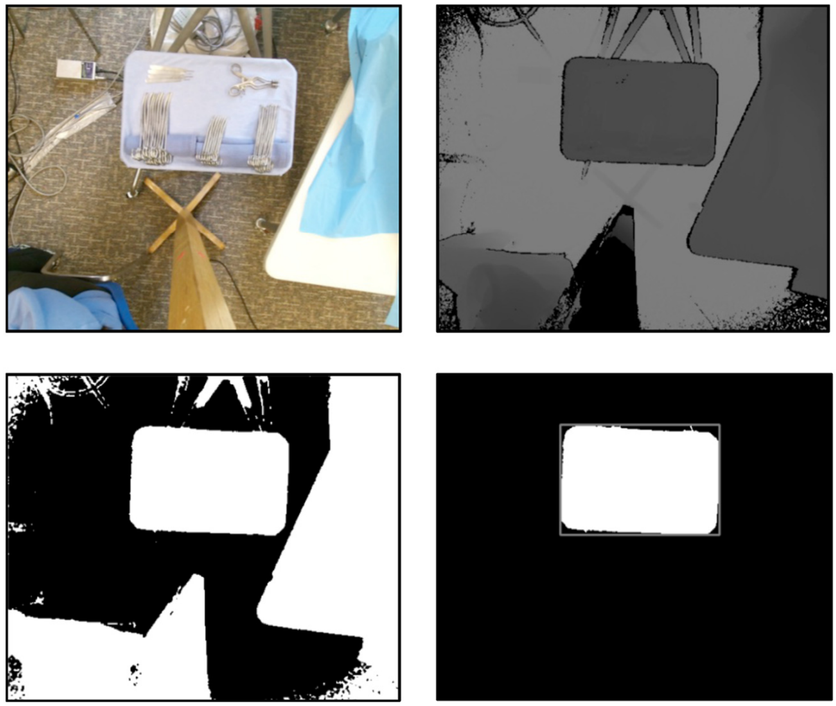

Fig. 4. Mayo segmentation procedure. (top left) raw color image I_C from Kinect; (top right) raw depth image I_D from Kinect visualized in grayscale; (bottom left) foreground mask M_{fg} generated from Otsu's threshold on I_D; (bottom right) mayo mask M_{mayo} generated from contour examination.

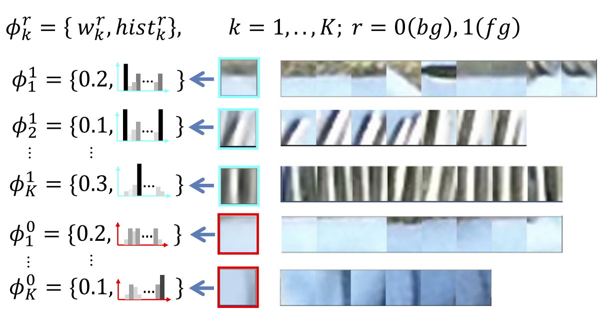

Fig. 5. Codebook clusters for foreground ϕ_k^1 (cyan) and background ϕ_k^0(red). Each codebook entry ϕ_k^r is a pair consisting of its importance (weight w_k^r) and the local color model (histogram $hist_k^r$). The patch with colored border (e.g. red, cyan) is the cluster's average image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

represents the relative importance of that local appearance. From a given cluster k, a 2D Hue-Saturation histogram (256 bins for each dimension) was calculated from the local color images of all the patches that belong to that cluster. This histogram from cluster k, denoted as $hist_k^r$, forms one of the K local color models for codebook ϕ^r, as shown in (1). The corresponding weight α_k^r of cluster k is used as the weight w_k^r for this local color model. An illustration of the generated codebook for the foreground and background categories is given in Fig. 5.

4.1.3. Segmentation

Given a color image I_C, the probability of each pixel belonging to a category c_r is estimated to generate a per-pixel segmentation. Such probability for a given pixel (x, y), denoted as $P(c_r|I_C(x, y))$, can be estimated using the Bayes minimum error criterion [36]. It requires the estimation of a priori probability $P(c_r)$ and a data likelihood $p(I_C(x, y)|c_r)$.

The prior $P(c_r)$ for each category is estimated using the relative frequency of all the patches associated with each category: $P(c_r) = \frac{|\{P_i|\theta_i = c_r\}|}{\sum_{s=1}^{R} |\{P_i|\theta_i = c_s\}|}$. The data likelihood $p(I_C(x, y)|c_r)$ was estimated through the marginalization over all codebook entries of that category:

$$p(I_C(x, y)|c_r) = \sum_{k=1}^{K} p\left(I_C(x, y), \phi_k^r|c_r\right)$$
$$= \sum_{k=1}^{K} P\left(\phi_k^r|c_r\right) p\left(I_C(x, y)|\phi_k^r, c_r\right) \tag{4}$$

The generated codebook as of Eq. (1) was used to estimate the parameters in the above equation. $P\left(\phi_k^r|c_r\right)$ is approximated using the weight w_k^r for the k_{th} codebook entry, indicating the relative importance of this codebook entry. $p\left(I_C(x, y)|\phi_k^r, c_r\right)$ is the likelihood of observing pixel $I_C(x, y)$ given category c_r and codebook entry ϕ_k^r. This likelihood was estimated using histogram back-projection (HBP). HBP algorithm can generate a probability of each pixel matching to a histogram from a given codebook entry. The segmentation probabilities calculated from HBP were accumulated using the corresponding weight w_k^r for the k_{th} codebook entry, according to as (4). Finally, a likelihood ratio between the different categories was calculated for each pixel in the image I_C. This value was used to segment foreground from background. The likelihood of a pixel (x, y) belonging to the foreground(c_1) w.r.t. the background (c_0) is:

$$L(x, y) = \frac{P(c_1|I_C(x, y))}{P(c_0|I_C(x, y))} \propto \frac{P(c_1)\, p(I_C(x, y)|c_1)}{P(c_0)\, p(I_C(x, y)|c_0)}$$
$$= \frac{P(c_1) \sum_{k=1}^{K} w_k^1 p\left(I_C(x, y)|hist_k^1\right)}{P(c_0) \sum_{k=1}^{K} w_k^0 p\left(I_C(x, y)|hist_k^0\right)}. \tag{5}$$

This value was compared with a threshold ρ to obtain a foreground mask (as $L(x, y) > \rho$). The hyper-parameter ρ was varied to generate a precision–recall curve in the later experiment.

4.1.4. Pose estimation

After the segmentation, the pose for each pack of unknown instrument was estimated for initial robot grasping, as shown

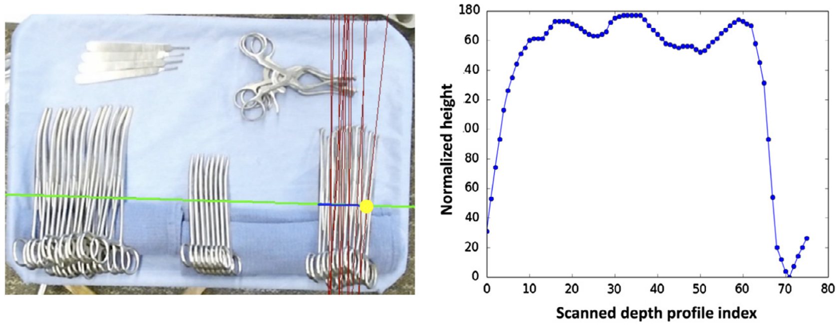

Fig. 6. Pose estimation process. (left) red lines are the result of Houghline transform and blue line indicates the perpendicular depth profile scanning line, the yellow point indicates the initial picking point. (right) the scanned depth profile. The fluctuation implies a vertical pose instead of flat pose. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in Fig. 6. The mayo tray is commonly organized so that all the instruments in each group are aligned in the same direction to avoid clutter. The cluster's direction is estimated and used to guide the grasping process for the tool. The dominant direction of each pile was found using the Hough transform [37] (Fig. 6 left). The lines extracted correspond to the edges of the instruments from a top-view. A majority vote of line directions was casted to find the dominant one. Estimating the instrument's group orientation is key to enable grasping of the individual instruments within the group. After the dominant direction has been identified, the line perpendicular to the dominant direction through the centroid of the pack was found (see the blue line in Fig. 6 left). The intersection of the line with the group contour was used as the initial grasping point (yellow dot in Fig. 6 left). The grasping strategy relies on picking the instrument from the center of mass and in a perpendicular direction.

After the position of the initial grasping point has been identified, the rotation of the gripper was estimated. The depth image was used to determine whether the instrument is lying flat on the mayo tray, or vertically packed within a group of instruments. The depth along the perpendicular line (blue) was scanned and its depth fluctuation was used to determine which of those two poses the instrument is at (Fig. 6 right). The standard deviation of the normalized height along the scanning line was calculated as: $\sigma_h = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(h_i-\mu)^2}$, where h_i is the depth value at point i (in total N points) and $\mu = \frac{1}{N}\sum_{i=1}^{N}h_i$ is the average. σ_h indicates the depth fluctuation of the instrument and is compared with an empirical threshold σ_{ref} to determine whether the instrument's state is flat or standing. The threshold σ_{ref} was chosen optimally based on a separate validation dataset. In the case where the instrument is lying flat, the robot would pick it from above; while a tilted instrument would require approach in a perpendicular direction to the instrument surface.

4.2. Reactive grasping protocol

Our approach to instrument recognition is done through manipulation of the instrument. Classical grasping approaches require a 3D model of the object that is to be grasped. Such approaches cannot be applied directly to our scenario since direct recognition of the instruments is not attainable. We first pick generic objects and only after manipulation will their category be determined. Assume that the instruments are packed in piles. The instrument orientation is determined through the value of the force feedback at the end-effector of the robot when in contact with the instrument. Once the orientation is determined, the instruments are picked by extending the end-effector in a perpendicular direction. A special end-effector was developed with an integrated electromagnet to attract metal instruments. Four force sensors were mounted between the electro-magnet gripper and the wrist of the robot, forming a cross shape in the xy-plane to acquire force information at the robot's end-effector, as its diagram shown in Fig. 7.

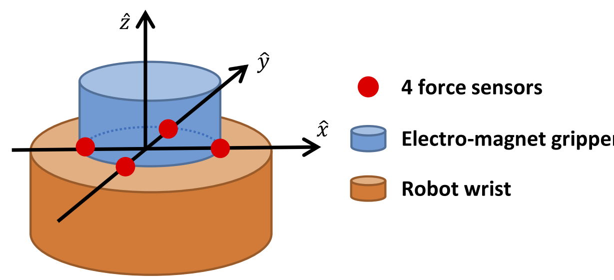

Fig. 7. Diagram of the force sensors and electromagnet. The four force sensors are placed on the x–y plane between the electromagnet and the robot wrist, forming a cross shape.

To improve the accuracy of force readings and compensate potential placement imbalances of the four force sensors, a calibration process was carried out before deployment. For raw force reading $\tilde{F}_x$(where $x \in \{up, down, left, right\}$), the minimum force $\tilde{F}_x^{min}$was recorded when the robotic arm is at a ready-to-pick position above the mayo stand, and the maximum force $\tilde{F}_x^{max}$ was recorded when the robotic arm is physically touching the instrument side ways to ensure that only force sensor x makes a point contact with the mayo stand. After recording the minimum and maximum readings for each sensor, the force value was normalized following equation:

$$F_x = \begin{cases} 0,\ if\,\tilde{F}_x < \tilde{F}_x^{min} \\ 1,\ if\,\tilde{F}_x > \tilde{F}_x^{max} \\ \dfrac{\tilde{F}_x - \tilde{F}_x^{min}}{\tilde{F}_x^{max} - \tilde{F}_x^{min}},\ otherwise \end{cases} \tag{6}$$

The resultant force F_x was in the range [0,1] and sampled at 5 Hz rate. The four normalized force readings are stacked as a vector $\vec{F} = \left[F_{up}, F_{down}, F_{left}, F_{right}\right]$. A sharp increase in the sensing force indicates contact between the end-effector and the instrument. Given the force readings $\vec{F}$ and the sensors' known locations under the electromagnet, the norm direction $\vec{n} = [n_1, n_2, n_3]^T$ of the force plane is estimated. When proper contact is being made, all the force sensors acquire approximately the same force value (the force is evenly distributed in the surface when the magnet is perfectly perpendicular to the plane of the instrument), thus resulting in a force norm vector $\vec{n}$ to be aligned with the z-axis $\hat{z}$. If the force norm is not aligned, then the end-effector's orientation is corrected towards the direction to reduce discrepancy between the norm

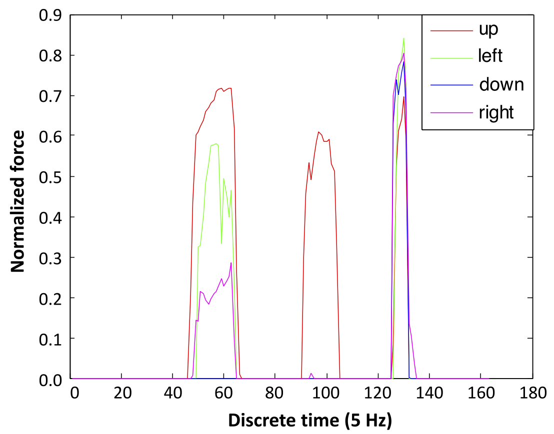

Fig. 8. Sample force readings for one instrument picking. The first peak is the first contact with instrument, while second peak indicates a second contact after adjusting orientation, and the last peak indicates a final stable contact (evenly distributed force).

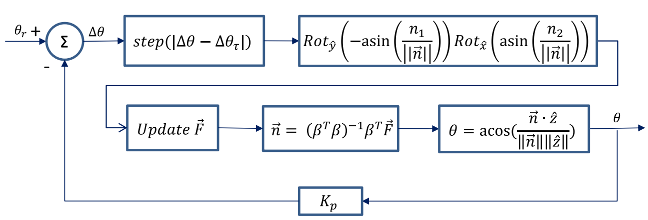

Fig. 9. System diagram for the reactive grasping protocol.

direction $\vec{n}$ and z-axis $\hat{z}$. Fig. 8 shows a sample force reading during one instrument picking procedure.

The norm direction $\vec{n}$ was estimated using least square regression with force input $\vec{F}$ following the equation: $\vec{n} = (\beta^T\beta)^{-1}\beta^T\vec{F}$, where β is the coefficient incorporating locations of force sensors in the wrist coordinate, denoted as $\beta = [1, 0, F_{up}; -1, 0, F_{down}; 0, 1, F_{left}; 0, -1, F_{right}] \in \mathfrak{R}^{4\times 3}$. The angle difference θ between $\vec{n}$ and z-axis $\hat{z}$ was calculated and compared against a reference angle θ_r to form the control input $\Delta\theta$. Here the reference angle θ_r was set to 0 to ensure an evenly distributed force during contact. Different θ_r can be chosen in favor of other grasping poses. A tolerance $\Delta\theta_\tau$ was used to limit the orientation correction. Once the absolute value of the angle discrepancy $|\Delta\theta|$ is less than the tolerance value $\Delta\theta_\tau$, the maximum electromagnet force is exerted to pick up the surgical instrument. While there is no guarantee that only one instrument is picked, this is the most likely scenario due to the magnitude of the force applied. The reactive grasping control algorithm is illustrated in Fig. 9.

4.3. Instrument recognition

Once the end-effector attracted an instrument, it was lifted and subsequently placed on a side surface (known as the recognition pad). The recognition pad has a uniform background and allows maximum exposure to enable instrument recognition. Then, an object recognition procedure was used to recognize the type of instrument. The instrument recognition framework is shown in Fig. 10. It includes foreground extraction, feature encoding and instrument classification. Detailed steps of each module are presented next.

The pipeline of the image processing module involves first to record an image frame I_{t-1} when no instruments are present in the recognition pad. A color-based background model BG_{t-1} is built

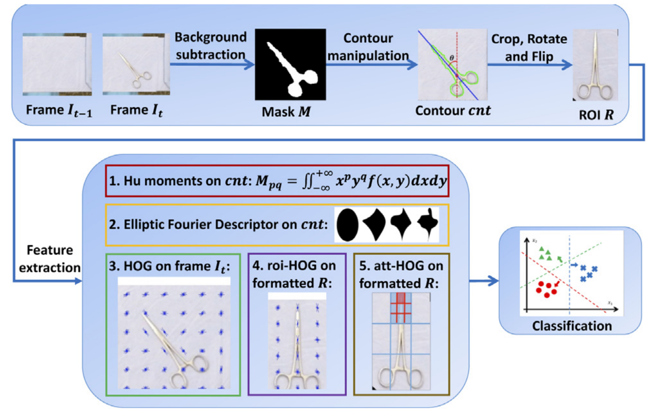

Fig. 10. Instrument recognition algorithm framework.

based on the pixel values of I_{t-1} using Gaussian Mixture Models [38]. Then, given the current frame I_t in which an instrument is present, the foreground mask M is extracted using I_t and BG_{t-1}. The object contour cnt is then extracted from foreground mask M. Certain region related properties are then extracted from the contour cnt to represent the foreground object.

Next, a straight line is fit to the contour pixels as the major axis for the object, and then the image of the foreground object is rotated based on the angle difference between the major axis and the vertical line. Thus, in the new rotated image, the foreground object looks vertically oriented. Last, the instrument image is flipped (if necessary) so that the lower half part of the object's area is larger than the upper half. The pixel area was used as an approximation for mass. This is a reasonable assumption for our scenario, since most surgical instruments have a scissors-like shape where the larger area corresponding to the handle has a larger mass. After localization, rotation and clipping, the Region-Of-Interest (ROI) image R which contains the foreground object was generated. Certain features were then extracted from R for instrument classification.

The features extracted from the Region-Of-Interest R were generated through a number of techniques described next. An attention-based HOG algorithm was developed (denoted as att-HOG), which has the capability of zooming in into specific sub-regions of an image to resolve ambiguity. The motivation behind the algorithm is that pairs of instruments have very similar appearances but only differ at a specific sub-region (i.e., hemostat and babcock forceps' tooltips). For such cases, features encoding this level of detail were created.

A visual illustration of the proposed attention-based HOG algorithm is given in Fig. 11. The algorithm works as follows. First, NROI-label pairs $\{(R_i, y_i) \,|i \le N, y_i \in G\}$ were given as input, where R_i is the ROI image (as shown in Fig. 10), y_i is the instrument label associated with R_i, and G is the set of all instrument labels. The algorithm follows an iterative structure with multiple scales (denoted by k). It starts with scale 0 ($k = 0$), where features were extracted from the full image of R_i. HOG features were extracted from R_i as a whole and a discriminative classifier clf_0 was trained to classify instruments: $R_i \xrightarrow{clf_0} y_i$. The performance of clf_0 was evaluated on a separate validation dataset and the average classification accuracy was recorded as s_0. Scale 0 is used as an initialization step and is not repeated any more. Then the algorithm moves to the next scale.

At scale 1 ($k = 1$), the entire image R_i was divided evenly into m rows and m columns, forming a total of m^2 patches. Patch j of image R_i was denoted as R_i^j. A discriminative classifier clf^j was trained from patch j (extracted from all the images) to classify instruments:

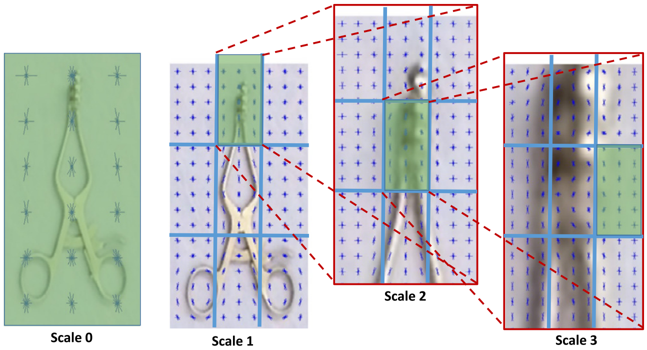

Fig. 11. Illustration of the attention-based HOG algorithm for instrument recognition. In scale 0, HOG features were extracted from the entire image at a fixed resolution. In scale 1, the entire image was divided into 3 × 3 patches (separated by blue lines) and HOG features were extracted from each grid (with fixed resolution). The patch yielding the best classification performance was chosen (highlighted with green) for the next scale. The selected patch was further divided into sub-patches, and the same process was iterated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$R_i^j \xrightarrow{clf^j} y_i$. The performance of clf^j was evaluated on a separate validation dataset and the average classification accuracy s^j was recorded. The same operation was repeated on each patch and in total m^2 trained classifiers and performance scores were generated at this scale, i.e. $\exists \left\{ \left(clf^j, s^j \right) | j \le m^2 \right\}$ for scale $k = 1$. The highest-scored classifier j^* was chosen following $j^* = arg\max_j s^j$. This classifier corresponds to the patch j^* which is the most discriminative to classify instruments at that scale. The selected best classifier clf^{j^*} and validation score s^{j^*} for scale k were further denoted as $clf_{(k)}$ and $s_{(k)}$. Then the algorithm moves to the next scale.

At scale 2 ($k = 2$), the input image R_i was replaced by the selected patch $R_i^{j^*}$ from the previous scale. The same procedure were then conducted with the selected patch instead of the entire image, so that attention is guided towards the more salient regions. This iterative process finishes when the iteration exceeds the maximum allowed number, or the size of the patch is less than a pre-set threshold.

When the iterative process is finished, a cumulative voting process is carried out to give the final classification result. Assume that the algorithm stops after K iterations, the best classifier and its corresponding validation score for scale k is $clf_{(k)}$ and $s_{(k)}$. Given an input R_i, the final decision $y_i^* \in G$ is made according to:

$$y_i^* = arg\max_{y \in G} \sum_{k=0}^{K} s_{(k)} P\left(y | clf_{(k)}, R_{(k)}\right) \quad (7)$$

where $P\left(y | clf_{(k)}, R_{(k)}\right)$ is the probability of classifying input $R_{(k)}$ into y using $clf_{(k)}$. This probability is calculated by the specific classifier.

To better evaluate the proposed instrument recognition algorithm, several algorithms were observed. They differ on the ways of generating features, namely:

- Hu – Hu moments [39] from contour *cnt*
- EFD—Elliptic Fourier Descriptor [40] from contour *cnt*
- HOG–HOG features [41] from the raw image I_t
- roi-HOG – HOG feature from ROI image R
- att-HOG – the proposed attention-based HOG features extracted from ROI image R

The generated features were then classified by several discriminative classifiers, including Support Vector Machines (with linear and radial basis function kernel) [42], Random Forest [43] and Adaboost [44]. The one-vs-all classification strategy was used to enable a multiclass recognition.

5. Experiments

Experiments were conducted to test the performance of the instrument segmentation, reactive grasping and instrument recognition algorithms, respectively. The image dataset used in our paper is publicly available at https://github.com/tian-zhou/Surgical-Instrument-Dataset.

5.1. Instrument pack segmentation experiment

This experiment aims to evaluate the proposed segmentation algorithm to discriminate the group of surgical instruments (foreground) from the mayo stand image (background). A dataset of surgical instrument images on the mayo stand was collected using Kinect, with various instrument layouts and lighting conditions. The instrument ground truth mask was manually annotated using the LabelMe toolbox [45]. Initially 60 image sets (color image, depth image and ground truth mask) were recorded and then each image was distorted by rotating in 8 angles (0 to 360 stepped by 45) to generate a larger dataset (resulting in total 480 images).

The experiment follows a 10-fold cross-validation setup, where in each fold, 90% of the entire dataset was used for training and the remaining 10% for testing. From each image in the training data, 100 random patches ($N = 100$) of size 16 × 16 ($w = 16$)were extracted for both foreground and background. The extracted patches from all the training images were then clustered by fitting a GMM model on their associated feature representations with 10 components ($K = 10$). Codebooks were then generated by saving the color histograms and corresponding weights for each cluster. For both foreground and background categories, a codebook was generated during the training process. For segmentation purposes, the weighted histogram backprojection algorithm was used to calculate the likelihood of a given pixel belonging to the foreground as opposed to the background. The likelihood ratio was then compared to the threshold ρ to determine the final segmentation result. The segmentation result was compared against the ground truth mask. The F1 score was used for evaluation, which is a common metric to evaluate segmentation result [46]. The likelihood threshold ρ was swept to obtain the precision–recall (PR) curve demonstrating the segmentation performance. The Area Under Curve (AUC) is used as a cumulative metric for evaluation. We compared the proposed segmentation algorithm against *grabCut* algorithm [23] and Morphological Geodesic Active Contours *(snake)* algorithm [47]. Both benchmark algorithms need an initial segmentation marker to start the full segmentation process. We generate the marker image using ground truth labels, where the centroid pixel of each instrument pile was labeled as positive. The segmentation was then carried out on the RGB image of the instrument. Fig. 12 presents the PR curves for the three different algorithms. The maximum F1 score achieved by each algorithm was also marked.

It is found that the proposed segmentation algorithm achieves the highest AUC and maxF1 score, followed by *grabCut* and last *snake* algorithm. It is also worth noting that the proposed algorithm does not require an initial marker to start the segmentation process, while both of the benchmark algorithms do.

5.2. Reactive grasping experiment

To assess the effectiveness of the grasping method, two methods were compared: one using the reactive grasping protocol (RG), and the other using the open-loop grasping (OLG) protocol. Both protocols use the estimated pose from the optical sensor to build an initial picking strategy, while the RG protocol additionally used force sensors to correct the orientation of the tool-tip. This experiment compared the instrument picking success rate of these two

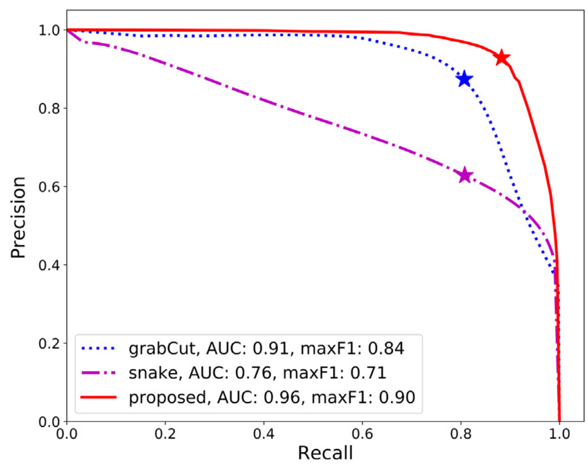

Fig. 12. Precision–recall curve for the proposed segmentation algorithm, compared against grabcut and active contour model (snake) algorithms. The AUC and max F1 score were shown in the legend. The max F1 scores were marked with star.

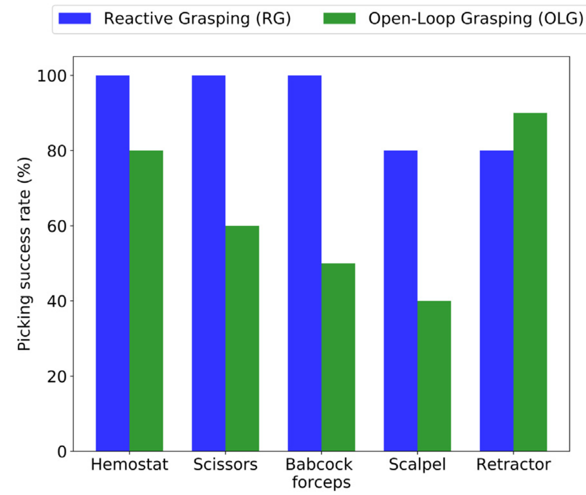

Fig. 14. Picking success rate for RG vs OLG.

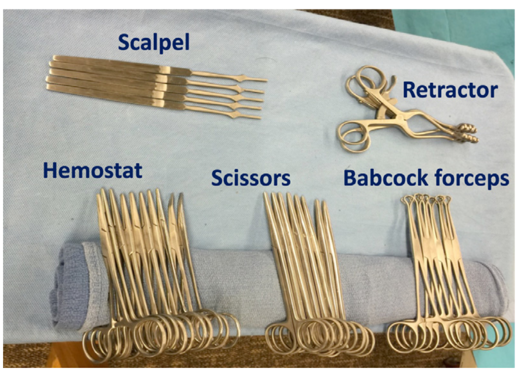

Fig. 13. Mayo setup for the grasping experiment. There are five sets of surgical instruments, with their name shown.

protocols. A seven degree of freedom (DOF) Barrett WAM robot arm was used in this experiment. Fig. 13 shows the setup for the mayo stand in this experiment. It includes five sets of surgical instruments, named retractor (2 pieces), scalpel (5), Babcock forceps (6), scissors (8) and hemostat (10). They were all grouped together in a way similar to the surgical setting in the operating room. For both grasping methods, the task is to pick up each surgical instrument in sequence successfully. The grasping process was repeated 10 times for each type of instrument, resulting in a total of 50 picks. The average picking success rate was 64% for OLG protocol and 92% for RG protocol. The per-instrument average picking success rate was shown in Fig. 14. A paired sample t-test between the per-instrument average success rates of the two protocols yield a one tail p-value of 0.029, which indicates the superiority of the RG policy compared to the OLG policy.

Since the instruments are packed tightly, the electromagnetic force was propagated to near-by instruments, resulting in potentially picking up multiple instrument at the same time. This was not a desired outcome and resulted in false alarms. Under the RG protocol, 6 out of the 46 successful picks include picking multiple instruments, resulting in a false alarm rate of 13%. The same number calculated for OLG protocol was 6.25%. The RG protocol has a higher false alarm rate, since the orientation correction algorithm leads to a better contact between the magnet and the instruments. Such better contact resulted in a stronger electromagnetic force propagated to the instrument pile. 62.5% of all the false alarms in both grasping methods occurred when picking the scalpel, since it is one of the lightest instruments. Five out of the twelve scalpel picks (with both grasping methods) resulted in false alarms, leading to a false alarm rate of 41.6% for the scalpel.

5.3. Instrument recognition experiment

A dataset of surgical instruments on the recognition pad was collected using our system setup. There are in total 5 types of instruments, named scalpel, retractor, scissors, hemostat and forceps. For each type of instrument, 20 images were taken initially with different illumination and layout conditions.

The dataset was increased by using label-preserving transformation [48] to produce variants of the original images. This helps to increase dataset size and also prevent over-fitting. Two distinct forms of data augmentation techniques were applied. The first consists of applying rotation and mirroring (horizontally) on the original image. Each input image was rotated in 8 angles (0 to 360 degrees stepped by 45), and then mirrored horizontally. The second consists of altering the RGB channel intensities. This technique follows an important property of natural images since an object recognition algorithm should be invariant to intensity and illumination changes. We followed the steps of [48]. A Principal Component Analysis (PCA) was first applied on the set of RGB pixel values of the entire training dataset. Then for each training image, multiples of the principal components were added. The magnitudes were proportional to the corresponding eigenvalues times a random variable drawn from a Gaussian with zero mean and standard deviation of 0.1. Therefore to each RGB image pixel $I_{xy} = \left[I^R_{xy}, I^G_{xy}, I^B_{xy}\right]^T$, we added the following brightness values:

$$[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3]\,[\alpha_1\lambda_1, \alpha_2\lambda_2, \alpha_3\lambda_3]^T \tag{8}$$

Where $\mathbf{p_i}$ and λ_i are the *ith* eigenvector and eigenvalue of the 3×3 covariance matrix of RGB pixel values, respectively, and α_i is the random variable mentioned before. Both the original version and the altered version were kept in the dataset. After applying both data augmentation techniques, 32 augmented images were generated out of 1 training seed, and the final dataset includes 3200 images with 640 for each instrument type. The learning setting follows a special 5-fold cross validation, where the original training instance and its augmented variants always reside in the same data split. This is to avoid the case where the original example was used for training and its variants were used for testing.

The hyper-parameters for each feature extraction method were chosen based on a grid search. The parameters yielding the highest

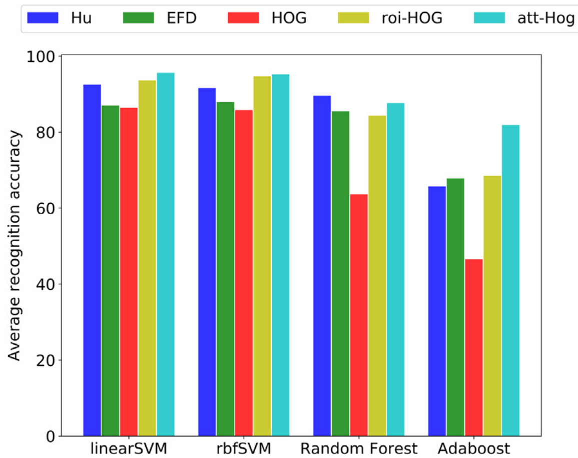

Fig. 15. The average recognition accuracy for different features using different classifiers. The best performance was achieved by attention-HOG with linear SVM, with a recognition accuracy of 95.6%.

Table 1
Average recognition accuracy for different features using different classifiers. The bottom row is the per-feature average accuracy and the right-most column is the per-classifier average accuracy.

	Hu	EFD	HOG	roi-HOG	att-HOG	avg_clf
linear SVM	92.5	87.0	86.4	93.6	**95.6**	91.0
rbf SVM	91.6	87.9	85.8	94.7	95.2	91.0
Random forest	89.6	85.5	63.6	84.3	87.7	82.1
Adaboost	65.7	67.8	46.5	68.5	81.9	66.1
avg_fea	84.8	82.1	70.6	85.3	90.1	NA

recognition accuracy on a separated split were chosen as the optimal parameters. In the HOG feature method, image I_t was resized to dimension 72×56, and then 8×8 cell size, 2×2 cells per block and 9 orientation bins were used for HOG calculation. In the EFD case, the first 5 orders of Fourier coefficients were used as features. For the roi-HOG case, the ROI image R was resized to 24×48, and then 8×8 cell size, 2×2 cells per block and 9 orientation bins were used. The att-HOG feature utilizes the same HOG configurations as roi-HOG, with a 3×3 grid split of each image ($m = 3$) and a maximum of 3 iterations ($K = 3$). For all feature extraction methods, the generated features were normalized to have mean of zero and standard deviation of one for each channel.

The features were classified using four classifiers, including linear SVM, rbf-kernel SVM, Random Forest and Adaboost. The resultant recognition accuracy on the testing split is shown in Fig. 15. In all the cases, the att-HOG features outperformed the roi-HOG features, which indicates the superiority of the proposed attention-based feature extraction structure.

The recognition accuracy for each feature-classifier combination is given in Table 1. The bottom row (i.e., avg_fea) shows the per-feature average and the right-most column (i.e., avg_clf) shows the per-classifier average. As shown, att-HOG is the best feature extraction method with the highest per-feature accuracy of 90.1% (followed by roi-HOG of 85.3%). On a different perspective, the best classifiers are linear SVM and rbf SVM with a tied per-classifier accuracy of 91% (followed by random forest of 82.1%). The best overall recognition accuracy was achieved by the proposed att-HOG algorithm with linear SVM, with an average accuracy of 95.6% (followed by att-HOG + rbf SVM with 95.2%). Recognition results showed that the strategy consisting of grasping and moving the instruments allows for reliable instrument recognition.

6. Discussion

The aim of this work was to tackle the problem of surgical instrument recognition based on visual perception and manipulation in a fashion that resembles human recognition. This is a challenging problem because the instruments look similar, are occluded and grouped together in clusters. To tackle this problem, we used a standard setup often found at the operating room. The experiment setup consisted of a simple yet common mayo tray with instruments organized in tight groups. It was found that the recognition accuracy reached 95.6% which is a satisfactory performance. One limitation of our work is that grasping might fail if too much overlap exists between the instruments. This could be attributed to an inaccurate estimate of the instruments' centroid. When picking the instrument from a location different than its centroid, the force applied may not be enough to grasp the object.

Regarding the instrument recognition algorithm, the current focus was to distinguish instruments of different types. This is a first and important step in developing a more complete instrument recognition module, where other variations such as different sizes, materials, and designs are considered. The current recognition framework provides the flexibility to add extra features for those purposes. For example, the volume of the instrument area can be used to distinguish different sizes of the same instrument, as occurs with scissors of different type (e.g. stitch scissors, tenotomy scissors, and Metzenbaum scissors).

Another potential approach for instrument recognition is to directly classify the instrument clusters. The main limitation with that approach is that the surgical instruments may look similar in such a setting, and are partially occluded by other instruments. Alternatively, instruments placed at a pad facing down would allow a more detailed examination, which may result in higher classification accuracies. Such insight is shown by the att-HOG' superior performance over roi-HOG. Computation-wise, extracting features from the entire instrument group would take more time compared to when analyzing an individual instrument's image (under the same feature resolution).

In terms of computational cost for the image processing algorithms the following was found. The segmentation procedure took in average 30ms to finish. When it comes to instrument recognition with linear SVM classifier (which showed the best performance), Hu, EFD, HOG, roi-HOG and att-HOG took 4 ms, 5 ms, 16 ms, 90 ms, 100 ms, respectively. The computational times were obtained when running the algorithms on an Intel Xeon CPU @ 3.2 GHz with 16 GB RAM, under Windows 10 64 bit environment. To summarize these findings, the proposed segmentation and recognition algorithms (att_HOG) are suitable for real-time performance.

7. Conclusions

In this paper we presented a solution to the challenge of accurate surgical instrument recognition by a Robotic Scrub Nurse in the Operating Room. Recognizing surgical instruments from mayo trays in a clinical setting is challenging due to the instrument clustered in groups, leading to self-occlusion, discontinuities and clutter. A hybrid computer vision and robotic manipulation strategy was adopted to tackle this challenge. Initially, the instruments were segmented and their poses estimated for robotic manipulation. Later, the RSN system picked up the unknown instrument and presented it in an optimal view for robust recognition. Experiments were conducted to evaluate the performance of each critical component of the system. The proposed segmentation algorithm achieved an F-score of 0.90, which outperforms the baseline *grabCut* algorithm (0.84) and *snake* algorithm (0.70). The proposed force-based grasping protocol achieved an average picking success rate of 92% with various instrument layouts, compared to a success rate of 64% for open-loop grasping protocol. The proposed attention-based instrument recognition module can reach a recognition accuracy of 95.6%, which is the highest among several benchmark methods. The experimental results proved the feasibility and effectiveness of the proposed RSN system.

The proposed hybrid perception and manipulation strategy can solve the surgical instrument recognition problem automatically, accurately and robustly. This is achieved by a mixture of computational algorithms and various hardware equipment. Each module is a necessary component in the entire framework, together achieves the final performance. The recognition-through-interaction paradigm is proven to be effective from the high recognition accuracy. This paper not only solves this specific surgical instrument recognition problem successfully, but also proposes innovative image processing algorithms (object segmentation and attention-based recognition), hardware designs and control strategies. These algorithms, designs and frameworks can shed light on other assistive robot scenarios, such as robots organizing parts/tools on the manufacturing floor to assist human workers, or robots recognizing and handing over daily objects to disabled human beings in a rehabilitation scenario.

Future work includes the design of an adaptive force controller to ensure one instrument being picked up every time. We also plan to deploy the developed system in the OR. To that end, it is necessary to design a safe instrument delivery protocol for a reliable human–machine collaboration in the surgical setting.

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